

Development of Custom Made Tunable Optical Spectrometer

PHYSICS PROJECT REPORT

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Objective of the Project

- To construct a simple optical spectrometer using a CD.
- To observe and analyse spectra of different light sources.
- To understand diffraction and interference practically.
- To explore tunability by changing geometry.

Why this project?

1. Problem with Conventional Spectrometer
2. Aim to Build a Low-Cost Alternative
3. Strong Physics Learning Value
4. Concept of "Tunable" System
5. Real-World Applications Connection
6. Skill Development (VERY IMPORTANT)

Theory used.

- ### Diffraction Principle.

  When light encounters a surface with very closely spaced lines (like a CD), it spreads out into different directions. This phenomenon is called diffraction.(3)
  When a beam of light interacts with a surface having very closely spaced structures, such as the tracks on a CD, it does not travel strictly in a straight line. Instead, it spreads into different directions. This spreading of light is known as diffraction.(5)

  - **Types of Diffraction:**
    Diffraction can be broadly classified into two types:
    Single-slit diffraction – occurs when light passes through a narrow slit and spreads out.
    Diffraction grating – occurs when light interacts with a large number of equally spaced slits or grooves.
    In this experiment, the CD acts as a reflection diffraction grating, where its closely spaced spiral tracks behave like multiple slits.

- ### Interference.

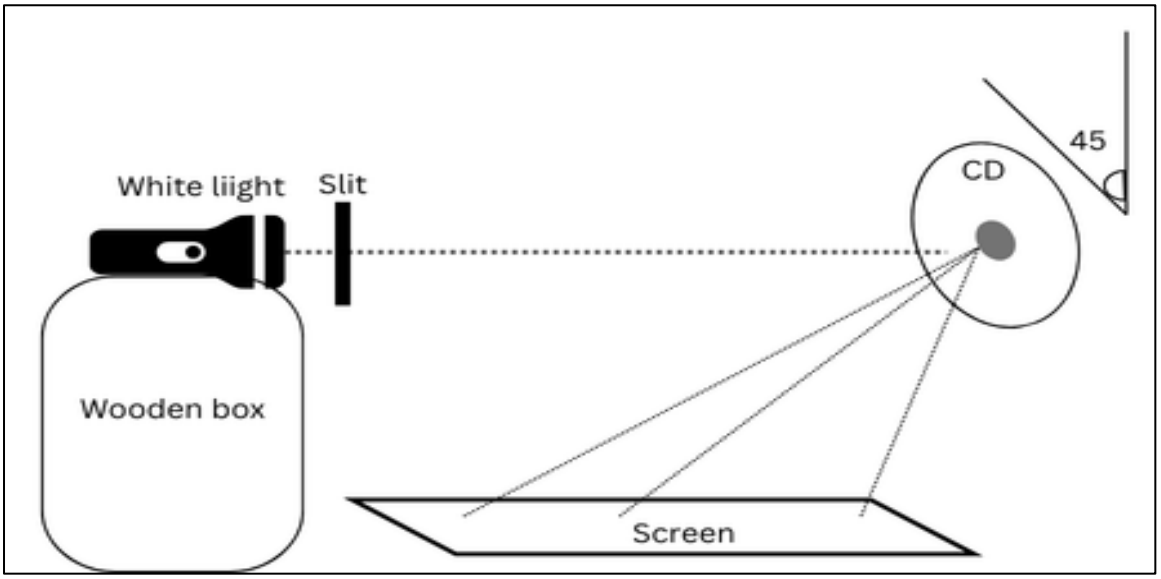
  Interference occurs when two or more coherent light waves overlap and combine, resulting in regions of constructive and destructive interference.(1) When light falls on the CD, each track reflects light waves.(4) These reflected waves from different tracks interfere with each other.

  - **Condition for interference:**
    For interference to occur, the following conditions must be satisfied:

- The light waves must be coherent (constant phase difference)
- They should have the same wavelength and frequency
- Their amplitudes should be comparable

- ## Grating Equation.

  The grating equation provides the condition under which constructive interference occurs for light diffracted by a diffraction grating. In this experiment, the CD acts as a **reflection diffraction grating**, where its closely spaced tracks behave like multiple slits.



$$d\sin\theta = n\lambda$$

(Derived from interference conditions in diffraction grating)(2)

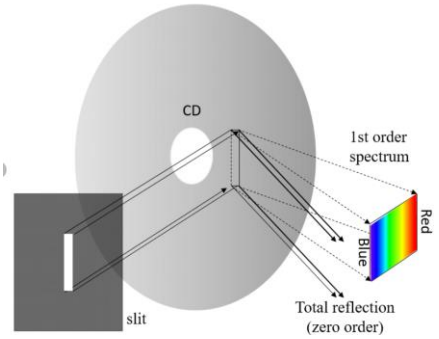
Role of Slit.

- Converts Extended Source → Narrow Beam
- Improves Resolution of Spectrum
- Prevents Overlapping of Spectral Lines
- Helps in Accurate Measurement
- Acts as a Spatial Filter



Why CD / DVD / VD works as grating?

- CD has thousands of closely spaced tracks
- These tracks act like multiple slits
- Causes interference and diffraction



Role of CCD / Mobile Camera.

For this project, a **mobile camera** is used as a detector in the spectrometer setup.

The mobile camera captures the diffracted spectrum formed after light interacts with the CD. It records the intensity distribution of light in the form of digital images, which can be further analysed using image processing software.

Unlike the human eye, which can only observe the spectrum qualitatively, the mobile camera allows the spectrum to be stored and examined in detail. The captured images can be used to extract intensity variations and position data (in pixels), enabling basic quantitative analysis.

Although a mobile camera is less sensitive compared to specialized detectors like CCD cameras, it provides a **low-cost and accessible alternative** for spectral observation. It enables visualization, documentation, and preliminary analysis of the spectrum.

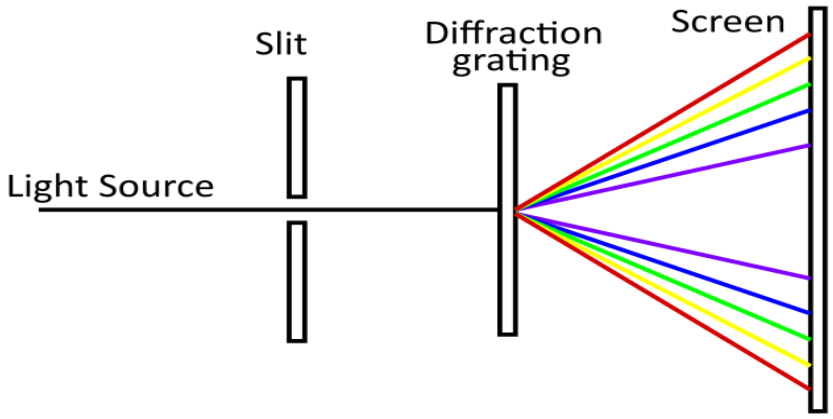
The use of a mobile camera makes the experiment more practical and portable while still allowing meaningful analysis of diffraction patterns.



Experimental Setup.

The performance of the optical spectrometer depends critically on the proper selection and alignment of its components. Each component in the setup has a specific role and must be positioned carefully to ensure accurate observation of the spectrum.

1. Overall Arrangement



📌 Arrangement Sequence: Light Source → Slit → CD (Grating) → Camera

2. Components & their placement:

A. Light Source:

The light source is placed at the beginning of the setup. We have used torch as a source of light. The primary function of the light source is to provide incident light containing multiple wavelengths. A white light source is preferred because it produces a continuous spectrum when diffracted. The intensity and uniformity of the light source directly affect the visibility of the spectrum.



B. Slit:

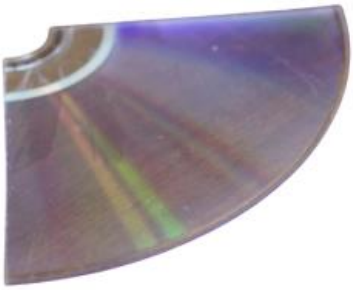
The slit is placed immediately in front of the light source.
Its role is to convert the light from an extended source into a narrow, well-defined beam. This is essential for achieving clear and sharp spectral lines.
The slit is aligned such that light passing through it travels in a straight path toward the CD.
The slit acts as a spatial filter and is one of the most critical components for improving resolution.



C. CD (Diffraction Grating):

The CD is placed after the slit and acts as a reflection diffraction grating. It is positioned at an angle of approximately **45°** with respect to the incident light beam. The reflective surface of the CD faces the slit. The CD is mounted on a

stable holder to maintain a fixed angle. Even a slight change in angle can significantly affect the position and clarity of the spectrum.



D. Camera:

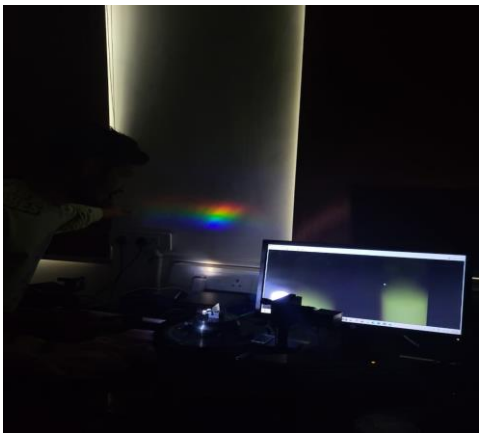
The camera is placed at a position where the diffracted light is visible. Instead of using a screen, a CCD / Mobile camera is used to capture the spectrum. This allows better visualization and the possibility of further digital analysis. The distance between the CD and the camera is typically maintained between **10–30 cm**. This distance allows the spectrum to spread sufficiently for clear observation. Proper positioning of the camera is essential for capturing sharp and measurable spectral images.

E. Supporting Structure (Optical Bench / Holders):

All components are mounted on an optical bench or stable holders to maintain alignment. Stability is crucial because even minor vibrations or displacement can disturb the setup and affect the results.

F. Dark Environment:

The entire experiment is conducted in a dark laboratory. This minimizes external light interference and enhances the visibility of the diffracted spectrum, which is usually faint. The quality of the observed spectrum improves significantly in a controlled dark environment.







Procedure.

The experiment was carried out in a controlled dark laboratory environment to minimize external light interference and enhance the visibility of the diffracted spectrum.

- **Step 1: Preparation of Experimental Setup**

All the components including the light source, slit, CD (diffraction grating), and CCD camera were arranged on an optical bench or stable platform. Care was taken to ensure that all components were properly fixed to avoid disturbances during the experiment.

- **Step 2: Adjustment of Light Source**

The light source (mobile flashlight/LED) was placed at a distance of approximately **5–15 cm from the slit**. The source was aligned such that light was directed straight toward the slit. This ensured that maximum light entered the slit for further processing.

Important Note:
The light source was kept sufficiently close to maintain intensity, but not too close to avoid uneven illumination.

- **Step 3: Slit Alignment and Adjustment**

The optical slit was placed directly in front of the light source. The width of the slit was carefully adjusted to obtain a narrow beam of light. Multiple trials were performed to find an optimal slit width.

- A wider slit produced a brighter but blurred spectrum
- A narrower slit produced a sharper but less intense spectrum

Key Observation:
The slit width significantly affected the clarity and resolution of the spectrum.

- **Step 4: Placement and Orientation of CD**

The CD was mounted on a holder and placed at a distance of approximately 2–10 cm from the slit. The reflective surface of the CD was oriented toward the slit. The CD was tilted at an angle of approximately 30°–45° with respect to the incident beam. The angle was adjusted carefully to ensure that the diffracted light could be observed clearly.

Important Note:
Even small changes in the angle of the CD resulted in noticeable changes in the position and clarity of the spectrum

- **Step 5: Alignment of Camera (Detector).**

A CCD camera was placed at a suitable position where the diffracted light from the CD could be captured. The distance between the CD and the camera was maintained in the range of 10–30 cm to allow sufficient spreading of the spectrum. The camera was aligned such that it directly faced the diffracted rays.

Important Note:
The camera position was adjusted multiple times to capture the maximum intensity and clarity of the spectrum.

- **Step 6: Dark Room Conditions**

The experiment was conducted in a dark laboratory to eliminate interference from external light sources. All unnecessary light sources were switched off, and only the experimental light source was used.

Key Insight:
A darker environment significantly improved the visibility of faint spectral lines.

- **Step 7: Fine Alignment and Optimization**

After initial setup, fine adjustments were made:

- The angle of the CD was varied slightly
- The slit width was adjusted
- The position of the camera was optimized

This step was repeated several times to obtain the clearest possible spectrum.

Important Insight:
Alignment was found to be the most critical factor affecting the success of the experiment.

- **Step 8: Observation of Spectrum.**

Once proper alignment was achieved, the diffracted spectrum was observed. The colors appeared in the order from violet to red. The intensity and clarity of the spectrum were noted.

- **Step 9: Image Capture**

The CCD camera was used to capture images of the observed spectrum.

Multiple images were taken under different conditions to compare results and improve clarity.

- **Step 10: Recording Observations**

All observations, including changes in slit width, CD angle, and camera position, were noted systematically.

These observations were used to analyze the performance of the spectrometer and identify limitations.

Working of Experiment.

The working of the spectrometer is based on the principle of diffraction and interference.

When light from the source passes through the slit, it forms a narrow beam. This beam falls on the CD, which acts as a reflection diffraction grating. The closely spaced tracks on the CD cause the light to diffract into different directions.

Different wavelengths of light are diffracted at different angles, resulting in the formation of a spectrum. The CCD camera captures this spectrum, allowing observation and further analysis.



Figure 1: AI generated Image

Quantitative Analysis.

In addition to qualitative observation of the spectrum, the spectrometer setup can be used to estimate the wavelength of light using diffraction principles and geometric measurements.

To estimate the wavelength of light, the following relation is used: - $$\tan\theta = \frac{x}{L}$$

For small angles: - $$\sin\theta \approx \frac{x}{L}$$ $$\lambda = \frac{d \cdot x}{nL}$$

Substituting into the grating equation: -

Where:

- x= distance of spectral line
- L= distance between CD and camera

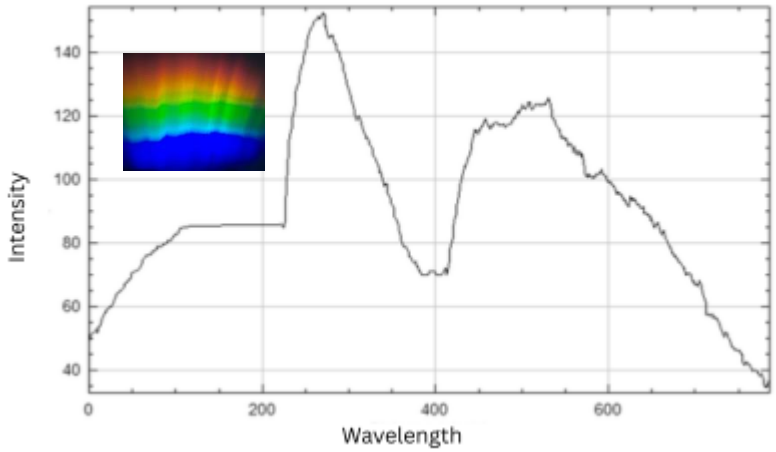
Sr. no	Wavelength(nm)	Intensity	Observation
1	~420 nm	~35	Very low intensity (Violet region)
2	~480 nm	~80	Increasing intensity (Blue region)
3	~520 nm	~120	First peak (Green region)
4	~560 nm	~110	Slight decrease
5	~590 nm	~70	Minimum intensity (Dip region)

Graphical Analysis.

To better understand the effect of the slit on the observed spectrum, intensity graphs were plotted under two conditions:

- With slit
- Without slit

(A) Graph with Slit.



The graph represents the variation of intensity with respect to wavelength for the spectrum obtained using a slit.

- ***Observations from the Graph***

A sharp and prominent peak is observed around ~250 nm (or corresponding wavelength region) with intensity reaching approximately 150 units, indicating strong constructive interference. The intensity shows a sudden rise before this peak, unlike the gradual variation observed without a slit.

A clear dip is observed around ~380–400 nm, followed by another rise in intensity. A secondary broad peak is observed in the region of ~500–550 nm, showing another dominant wavelength region.

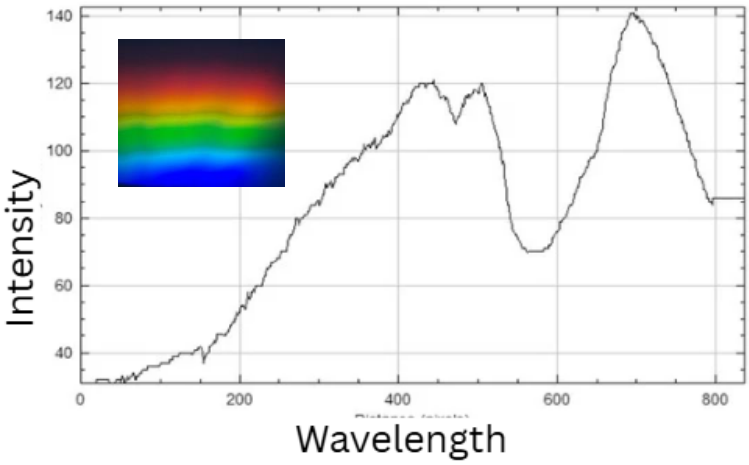
The curve shows more distinct variations and sharper features compared to the smooth curve obtained without slit.

- ***Conclusion from Graph***

The graph clearly demonstrates that the use of a slit significantly improves the quality of the spectrum. The presence of sharper peaks and distinct intensity variations indicates better resolution and separation of wavelengths.

This confirms that the slit is essential for obtaining accurate and well-defined spectral measurements.

(B) Graph without Slit.



The graph represents the variation of intensity (gray value) with respect to position (pixel distance) for the spectrum obtained without using a slit.

- ***Observations from the Graph***

• The intensity increases gradually from the initial region (~0–300 pixels), indicating a smooth distribution of light rather than sharp separation.
• Multiple broad peaks are observed in the range of approximately 400–750 pixels, but these peaks are not sharply defined.
• A noticeable dip in intensity is observed around 550–600 pixels, followed by a rise to a major peak near 700 pixels.

• The overall curve is smooth and continuous, without sharp spikes, indicating overlapping of spectral components.

- ***Conclusion from Graph***

The graph clearly shows that without a slit, the spectrometer produces a **diffused and poorly resolved spectrum**. The overlapping of wavelengths results in broad intensity distributions, making it difficult to identify individual spectral components accurately.

This confirms that the slit is essential for obtaining a sharp and well-defined spectrum.

(C) Comparison and Analysis.

- ***Graph with slit***

• The intensity shows sharp and well-defined peaks.
• Distinct features such as peaks and dips are clearly visible.
• Overlapping of wavelengths is significantly reduced.
• The spectrum shows better separation of different wavelength regions.

- Spectral resolution is high, allowing clearer identification of intensity variations.

➢ ***Graph without slit***

- The intensity varies smoothly across the spectrum.
- The peaks are broad and not sharply defined.
- Significant overlapping of wavelengths is observed.
- The curve appears continuous with gradual changes in intensity.
- Spectral resolution is poor, making it difficult to distinguish individual wavelength regions clearly.

Challenges / Constraints.

During the experiment, the following challenges were encountered:

- Difficulty in achieving precise alignment of components
- Low intensity of the diffracted spectrum
- Sensitivity of results to slit width
- Difficulty in capturing clear images using the camera
- Lack of accurate method for measuring diffraction angle

Key Insight:
Most challenges were experimental rather than theoretical.

Discussions.

The experiment demonstrates that a CD can effectively act as a diffraction grating. However, the quality of the observed spectrum depends strongly on alignment, slit width, and light intensity.

The limitations observed indicate that experimental precision plays a crucial role in optical measurements.

Further improvements in setup and calibration are required to obtain reliable quantitative results.

Conclusion.

The project successfully demonstrates the basic working of a spectrometer using a CD as a diffraction grating. A visible spectrum was observed, confirming the applicability of diffraction principles.

Although the setup is functional, further improvements are required for accurate measurement and analysis. This project provides valuable insight into experimental optics and instrument design.

Refrences.

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